ROAd: Robotaxi Optimized Allocation

Requirements Analysis and Specification Document

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Chapter 1

Introduction

1.1 Purpose

1.1.1 General purpose

ROAd, Robotaxi Optimized Allocation, is a software system designed to support the management of an autonomous taxi fleet in an urban environment. The system aims to help fleet operators monitor robotaxis, handle passenger ride requests, allocate vehicles efficiently, and rebalance idle vehicles across the city according to expected demand.

The main objective of ROAd is to reduce passenger waiting time and improve the overall efficiency of the fleet. To achieve this, the system supports different vehicle allocation strategies, such as assigning the nearest available robotaxi or selecting the vehicle with the minimum estimated arrival time. The selected strategy can be changed by the fleet operator at runtime, allowing the system to adapt to different operational needs. Moreover, this is enforced by a dynamic fleet rebalancing, repositioning ilde taxis towards areas with a high demand according to historical and real-time data.

ROAd also supports advance ride booking, requiring the system to manage vehicle availability over time and avoid conflicts between immediate and scheduled rides. Moreover, the system explicitly tracks the lifecycle of each robotaxi, from availability to assignment, arrival, ride execution, and maintenance. Passengers are notified about relevant changes in the state of their assigned vehicle.

The project focuses on the software architecture and management logic of the robotaxi allocation platform. Autonomous driving, payment processing, and mapping services are considered external dependencies and are not implemented as part of the system.

1.1.2 Goals

- [**G1**] Users must be able to register, log in, and manage their personal account.
- [**G2**] Users must be able to request an immediate robotaxi ride by providing pickup and destination locations.
- [**G3**] Users must be able to book a robotaxi ride in advance by specifying pickup location, destination, date, and time.

- [**G4**] The system must be able to assign an available robotaxi to a ride request according to the allocation strategy selected by the fleet operator.
- [**G5**] The fleet operator must be able to select and change the vehicle allocation strategy at runtime.
- [**G6**] The system must be able to track and update the state of each robotaxi during its lifecycle.
- [**G7**] Users must be notified about relevant updates concerning their ride, such as vehicle assignment, vehicle arrival, and ride status changes.
- [**G8**] The fleet operator must be able to monitor the current status and position of the robotaxi fleet.
- [**G9**] The system must be able to identify zones with expected high demand and suggest or trigger the rebalancing of idle robotaxis.
- [**G10**] The system must prevent conflicting assignments of the same robotaxi to overlapping rides or bookings.

1.2 Scope

1.2.1 General Scope

The S2B will interact with both passengers and fleet operators. To do so, it will offer to the former a mobile application, through which they can request or book robotaxi rides, and to the latter a web graphical user interface, through which they can monitor and manage the autonomous fleet.
The main feature offered to passengers will be to request a robotaxi ride directly from the application, either immediately or in advance, reducing waiting times and simplifying urban transportation.
Fleet operators will have the role of supervising the robotaxi fleet, selecting the vehicle allocation strategy, monitoring vehicle states and positions, and managing the distribution of idle vehicles across the city. The system will also support demand-driven rebalancing, helping operators improve fleet efficiency and reduce passenger waiting time.

1.2.2 Phenomena

According to the distinction between world phenomena, machine phenomena, and shared phenomena, as illustrated in the paper "The World and the Machine" by M. Jackson and P. Zave, the ROAd system involves several interactions between passengers, fleet operators, robotaxis, and external services. World phenomena are controlled by the external environment, machine phenomena are controlled by the system, while shared phenomena are observable by both the system and the external world.

Phenomenon	**Controller**	**Shared**
Passenger decides to use the ROAd service	W	N
Passenger registration	W	Y
Passenger login	W	Y
Passenger submits an immediate ride request	W	Y

Phenomenon	Controller	Shared
Passenger submits an advance ride booking request	W	Y
Passenger provides pickup and destination locations	W	Y
Passenger cancels a pending or scheduled ride	W	Y
Fleet operator login	W	Y
Fleet operator selects the vehicle allocation strategy	W	Y
Fleet operator monitors fleet status and positions	W	Y
Robotaxi sends its current position to the system	W	Y
Robotaxi sends its current operational state to the system	W	Y
Robotaxi is moved toward a rebalancing zone	W	Y
Robotaxi enters maintenance	W	Y
External mapping service provides route and travel-time information	W	Y
External demand data source provides historical or real-time demand data	W	Y
Ride starts when the passenger boards the assigned robotaxi	W	Y
Ride ends when the passenger reaches the destination	W	Y
System validates user credentials	M	N
System stores passenger and fleet operator account data	M	N
System checks robotaxi availability	M	N
System checks for conflicts between immediate rides and advance bookings	M	N
System assigns a robotaxi to a ride request	M	Y
System applies the currently selected allocation strategy	M	N
System updates the state of a robotaxi	M	Y
System updates the state of a ride	M	Y
System marks a robotaxi as unavailable during maintenance	M	Y
System notifies the passenger about vehicle assignment	M	Y
System notifies the passenger about robotaxi arrival	M	Y
System notifies the passenger about ride status changes	M	Y
System identifies zones with expected high demand	M	N

Phenomenon	Controller	Shared
System suggests or triggers the rebalancing of idle robotaxis	M	Y

1.3 Definitions, Acronyms, Abbreviations

1.3.1 Definitions

- **User:** a generic actor who interacts with the ROAd system. Depending on the context, a user can be either a passenger or a fleet operator.
- **Passenger:** a person who uses the ROAd mobile application to request an immediate robotaxi ride, book a ride in advance, monitor the assigned vehicle, and receive ride-related notifications.
- **Fleet Operator:** a person responsible for supervising and managing the robotaxi fleet through the ROAd web interface. The fleet operator can monitor vehicles, select allocation strategies, and manage fleet rebalancing.
- **Robotaxi:** an autonomous taxi belonging to the fleet managed by ROAd. A robotaxi can be assigned to passengers, execute rides, become unavailable, or enter maintenance.
- **Fleet:** the set of robotaxis managed by the ROAd system in the considered urban area.
- **Ride Request:** a request submitted by a passenger to travel from a pickup location to a destination. A ride request can be immediate or scheduled in advance.
- **Immediate Ride:** a ride requested by a passenger to be served as soon as possible.
- **Advance Booking:** a ride request scheduled by a passenger for a future date and time. The system must reserve a suitable robotaxi while avoiding conflicts with other assignments.
- **Pickup Location:** the location where the passenger wants to be picked up by the assigned robotaxi.
- **Destination:** the location where the passenger wants to be dropped off.
- **Allocation Strategy:** a policy used by the system to select which available robotaxi should be assigned to a ride request. Examples include selecting the nearest available vehicle or the vehicle with the minimum estimated arrival time.
- **Runtime Strategy Selection:** the possibility for the fleet operator to change the active allocation strategy while the system is running, without stopping the service.
- **Vehicle State:** the current lifecycle condition of a robotaxi. In ROAd, relevant states include available, assigned, arriving, arrived, in ride, and maintenance.
- **Available Robotaxi:** a robotaxi that is currently not assigned to any ride, not executing a ride, and not under maintenance.
- **Assigned Robotaxi:** a robotaxi that has been selected by the system to serve a passenger's ride request.

- **Idle Robotaxi:** a robotaxi that is available and not currently serving a passenger. Idle robotaxis may be considered for fleet rebalancing.
- **Fleet Rebalancing:** the process of repositioning idle robotaxis toward areas where high demand is expected, in order to reduce future passenger waiting times.
- **Demand Zone:** an area of the city associated with expected or detected ride demand.
- **High-Demand Zone:** a demand zone where the system predicts or detects a high number of ride requests.
- **Estimated Time of Arrival:** the estimated time needed by a robotaxi to reach a passenger's pickup location.
- **Notification:** a message sent by the system to a passenger or fleet operator to communicate relevant events, such as vehicle assignment, vehicle arrival, ride status changes, or rebalancing information.
- **Maintenance:** a condition in which a robotaxi is unavailable because it requires inspection, repair, cleaning, or any other operation that prevents it from serving rides.
- **External Mapping Service:** an external service used by ROAd to obtain geographical information, routes, distances, and travel-time estimations.
- **External Demand Data Source:** an external or historical data source used by ROAd to estimate future ride demand in different city zones.
- **Conflict:** a situation in which the same robotaxi would be assigned to two or more overlapping rides or bookings. The system must prevent such situations.

1.3.2 Acronyms

- **ROAd:** Robotaxi Optimized Allocation.
- **S2B:** Software To Be, the software system described and specified in this document.
- **API:** Application Programming Interface.
- **ETA:** Estimated Time of Arrival.
- **GPS:** Global Positioning System.
- **UI:** User Interface.
- **GUI:** Graphical User Interface.
- **FSM:** Finite State Machine.
- **UML:** Unified Modeling Language.

1.3.3 Abbreviations

- **[Gn]:** used to identify the n-th goal.
- **[Rn]:** used to identify the n-th functional requirement.

- **[NFRn]:** used to identify the n-th non-functional requirement.
- **[Dn]:** used to identify the n-th domain assumption.
- **[UCn]:** used to identify the n-th use case.
- **[SDn]:** used to identify the n-th sequence diagram.
- **[FSMn]:** used to identify the n-th finite state machine.

1.4 Document Structure

- **Chapter 1 – Introduction:** this chapter introduces the ROAd system, describing its purpose, goals, scope, main phenomena, and terminology. It also provides the structure of the document.
- **Chapter 2 – Overall Description:** this chapter provides a general description of the system from a requirements perspective. It includes relevant scenarios, the domain model, user characteristics, main product functions, non-functional aspects, and the main assumptions, dependencies, and constraints.
- **Chapter 3 – Additional Models:** this chapter provides additional models used to clarify the expected behavior of the system. In particular, it includes requirements-level sequence diagrams and finite state machines for the most relevant system processes.
- **Chapter 4 – References:** this chapter lists the documents, papers, standards, and other sources used as references for the preparation of this document.

Chapter 2

Overall Description

2.1 Scenarios

This section provides an informal description of relevant scenarios to illustrate the shared phenomena between the ROAd system and the external world, highlighting the interactions involving the main actors: Passengers and Fleet Operators.

Scenario 1: Immediate Ride Request and Vehicle Lifecycle
Alice, a registered Passenger, needs to travel from her office to the city center. She opens the ROAd mobile application and submits an immediate ride request by providing her destination. The system processes the request and applies the currently active allocation strategy, assigning the nearest available robotaxi to Alice. The chosen robotaxi transitions from the available state to the assigned state, and Alice receives a notification on her app confirming the assignment. Shortly after, the robotaxi starts moving towards Alice's pickup location, transitioning to the arriving state, and Alice is notified again. Once the vehicle reaches the pickup location, it enters the arrived state, and Alice receives a final notification to board. Alice boards the robotaxi, changing its state to in-ride. Upon reaching the destination, Alice exits the vehicle, and the robotaxi becomes available again to serve new requests.

Scenario 2: Advance Ride Booking
Bob wants to ensure he arrives at the airport on time. He opens the ROAd application and books a robotaxi in advance, specifying his home as the pickup location, the airport as the destination, and scheduling the pickup for 7:00 AM the following day. The system registers the advance booking and verifies database constraints to ensure vehicle availability for that specific timeline, successfully avoiding conflicts with other scheduled rides. The next morning, shortly before the scheduled time, the system assigns a suitable robotaxi to Bob's request and notifies him as the vehicle transitions to the arriving state.

Scenario 3: Automated and Manual Allocation Strategy Management
Mark is a Fleet Operator supervising the city's robotaxi fleet through the ROAd web interface while the system is in Auto Mode. During a period of heavy rain, traffic congestion in the city center increases. The system detects the traffic level as Medium and displays a notification on Mark's dashboard suggesting a switch from the default Nearest Available Vehicle strategy to the Minimum ETA policy to mitigate delays. Mark observes the sit-

uation but decides to remain in Auto Mode. As congestion continues to rise, the system detects a High traffic level and automatically switches the active strategy to Minimum ETA to ensure faster passenger pickups. Later, Mark determines that a specific operational condition requires a manual override, he manually selects the Nearest Available strategy. The system immediately transitions to Manual Mode, suspending all automated strategy changes until Mark explicitly re-enables Auto Mode via the dashboard interface.

Scenario 4: Dynamic Fleet Rebalancing
The ROAd system continuously collects and analyzes historical and real-time data provided by external demand data sources. Late in the evening, a major concert ends at the city stadium. The system predicts this area as a high-demand zone. To minimize customer wait times, the system proactively initiates dynamic fleet rebalancing. It identifies several idle robotaxis scattered in low-demand areas of the city and triggers their repositioning toward the stadium. Mark, the Fleet Operator, observes these rebalancing movements on his monitor as the vehicles are routed to efficiently manage the impending surge in ride requests.

2.2 Domain Model

The domain model captures the core entities of the ROAd ecosystem and the relationships between them, representing the conceptual structure of the urban autonomous mobility service.

Main Entities:

- **User:** An abstract entity representing any person interacting with the system, holding account credentials and personal data.
- **Passenger:** A specialized user who utilizes the mobile application to request or book rides and receive notifications.
- **Fleet Operator:** A specialized user responsible for supervising the fleet, selecting allocation strategies, and managing rebalancing through the web interface.
- **Robotaxi:** An autonomous vehicle characterized by its unique ID, current GPS position, and operational state (Available, Assigned, Arriving, Arrived, In Ride, Maintenance).
- **Ride Request:** A conceptual entity representing a passenger's need for transport. It contains the pickup location, destination, and status. A request is initially pending, then it can be accepted if a suitable robotaxi is assigned, rejected if no feasible vehicle is available, cancelled by the passenger, or completed after the ride has ended.
  - **Immediate Ride:** A request to be served as soon as possible.
  - **Advance Booking:** A request scheduled for a specific future date and time.
- **Ride:** An actual transport service generated from an accepted Ride Request. It represents the execution of the trip assigned to a Robotaxi, from pickup to destination, and keeps track of its lifecycle through states such as Scheduled, Waiting for

Pickup, In Progress, Completed, or Cancelled.

- **Location:** A geographical point defined by coordinates, used to specify Pickup and Destination points.
- **Zone:** A partition of the urban area used to track demand and trigger fleet rebalancing.
- **System Mode:** An attribute of the ROAd system that determines the control logic for vehicle allocation, supporting Auto and Manual modes.
- **Traffic Level:** A dynamic parameter provided by external mapping services, classified as Low, Medium, or High, which serves as the input for automated strategy switching.
- **Allocation Strategy:** The policy used for vehicle assignment.
  - **Nearest Available Vehicle:** Optimized for energy efficiency and battery preservation (default).
  - **Minimum ETA:** Optimized for service performance and passenger wait-time reduction during congestion.
- **Maintenance Event:** An event representing a period in which a Robotaxi is unavailable due to inspection, repair, cleaning, or any other operational reason. During a maintenance event, the corresponding Robotaxi cannot be assigned to ride requests.

Key Relationships:

- **Passenger – Ride Request:** A Passenger can submit multiple Ride Requests over time, but each request is associated with exactly one Passenger.
- **Ride Request – Location:** Each Ride Request must specify exactly one Pickup Location and one Destination.
- **Passenger – Ride:** A Passenger can take multiple Rides over time, but each Ride is associated with exactly one Passenger.
- **Robotaxi – Ride Request:** An Available Robotaxi is assigned to a Ride Request by the system based on the active strategy. A Robotaxi can be assigned to only one ride at a time to prevent conflicts.
- **Fleet Operator – Allocation Strategy:** The Fleet Operator selects and can change the active Allocation Strategy at runtime.
- **Fleet Operator – System Mode:** The Fleet Operator can switch the system between Auto and Manual modes via the web interface.
- **System – Allocation Strategy:** In Auto Mode, the system dynamically selects the Allocation Strategy based on the current Traffic Level and provides suggestions to the operator.
- **Fleet Operator – Allocation Strategy:** In Manual Mode, the Operator directly selects the strategy; performing this action while in Auto Mode triggers an automatic override and switches the system to Manual Mode.

- **External Mapping Service – Traffic Level:** The external service provides real-time Traffic Level data used by the system to evaluate strategy transitions.
- **Robotaxi – Zone:** Every Robotaxi is located within a specific Zone at any given time. Zones are used to identify High-Demand areas for Rebalancing.
- **Fleet – Robotaxi:** A Fleet is composed of multiple Robotaxis managed by the system.
- **Ride Request – Ride:** An accepted Ride Request can generate one Ride, while rejected or cancelled requests do not produce any Ride.
- **Robotaxi – Maintenance Event:** A Robotaxi can be associated with multiple Maintenance Events over time. When a Maintenance Event is active, the Robotaxi is considered unavailable for assignment.

Class Diagrams

The diagrams presented in this section describe the domain model of the ROAd system. They provide a conceptual view of the main entities involved in the autonomous robotaxi allocation service, together with their relevant attributes and relationships. Since the purpose of these diagrams is to support requirements analysis, they abstract from implementation details and do not aim to represent the complete software architecture or all the classes that will be introduced in the design phase. Some attributes are represented through enumerations in order to make the possible states or categories explicit.

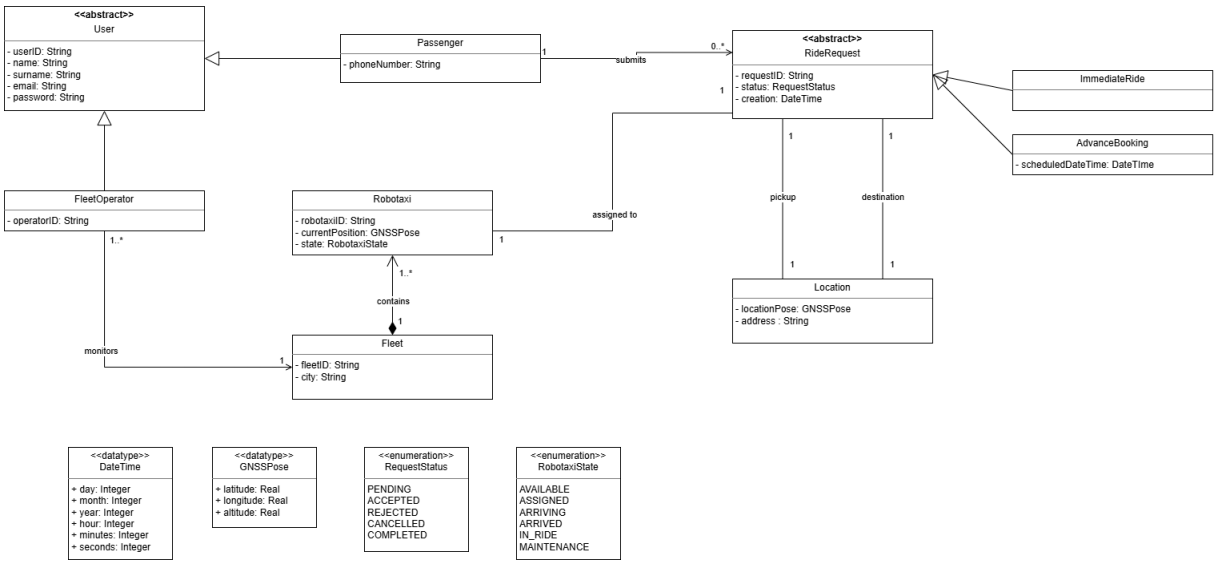

Figure 2.1: Core ROAd domain entities

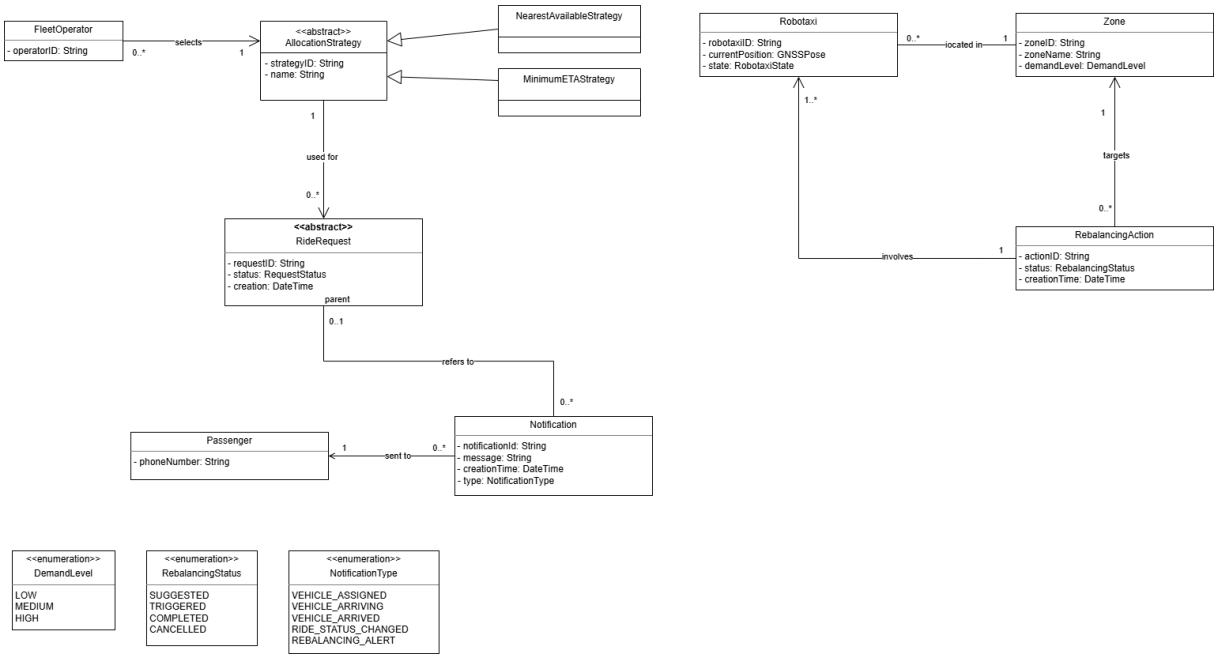

Figure 2.2: Allocation, rebalancing, and notification entities

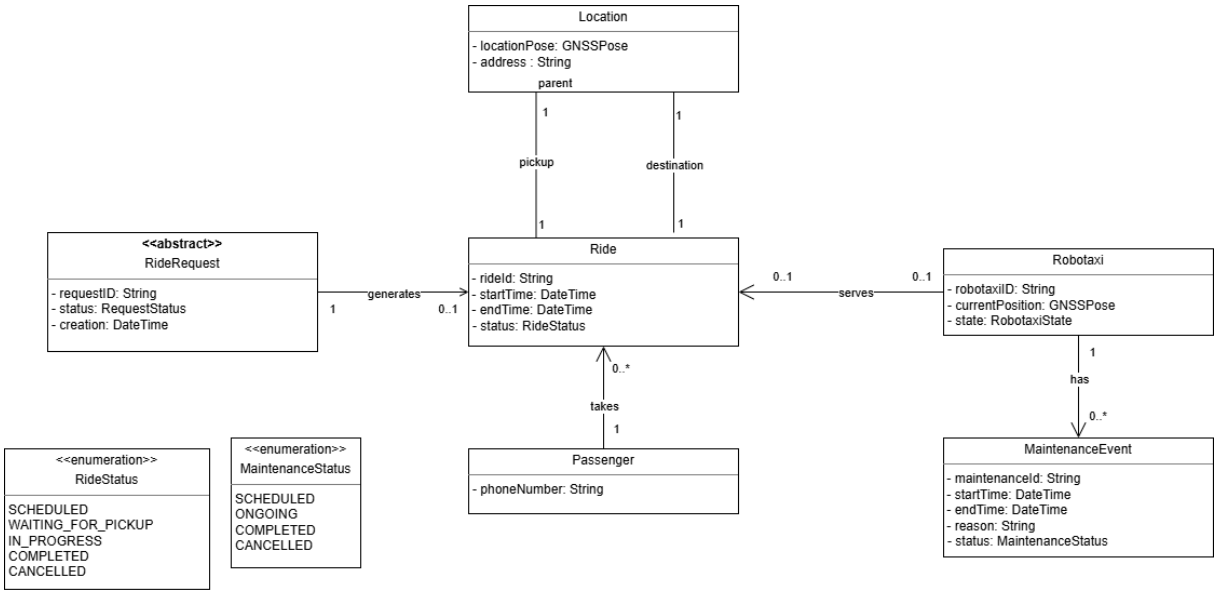

Figure 2.3: Ride execution and robotaxi lifecycle entities

Together, these diagrams provide a conceptual view of the ROAd domain and clarify the main entities that will be referenced by the requirements and additional models.

2.3 User Characteristics

The ROAd system is designed to be used by two primary categories of users, each with distinct roles, technical requirements, and operational goals. Understanding these char-

acteristics is essential to ensure the system meets the practical needs of urban mobility.

Passengers
Passengers represent the general public and are the primary consumers of the robotaxi service. Their main objective is to obtain reliable, point-to-point transportation with minimal effort and waiting time.

- **Technical Literacy:** Passengers are expected to have a basic to intermediate level of familiarity with mobile applications. The interface must be intuitive, requiring no specialized training.
- **Needs:** They require a streamlined process for requesting immediate rides or scheduling advance bookings. Real-time updates and clear notifications regarding vehicle status (e.g., assignment, arrival) are critical for a positive user experience.
- **Mobility Patterns:** Their usage varies from daily commuting to occasional trips (e.g., airport transfers), necessitating a system that can handle both spontaneous and planned requests efficiently.

Fleet Operators
Fleet Operators are professional users responsible for the supervision, management, and optimization of the autonomous vehicle fleet. They act as the "controllers" of the system's high-level logic.

- **Technical Literacy:** They are expected to be proficient in using web-based management interfaces and interpreting data visualizations (e.g., fleet maps, demand heatmaps). They must understand the implications of different vehicle allocation strategies.
- **Responsibilities:** Their primary tasks include monitoring the real-time distribution and state of the fleet and overseeing proactive rebalancing operations. Crucially, they act as supervisors of the automated allocation logic; they are responsible for evaluating system-generated suggestions triggered by traffic fluctuations and executing manual overrides to take direct control of the active allocation strategy during exceptional or unforeseen operational conditions.
- **Needs:** They require a comprehensive "command and control" dashboard that provides a clear overview of traffic levels, high-demand zones, and vehicle lifecycle transitions. Furthermore, the interface must clearly display the current operational state of the system (Auto or Manual mode) and provide immediate alerts for strategy switch suggestions, empowering them to make informed decisions safely and without service interruption.

2.4 Product Functions

The main functions of the ROAd system are categorized based on the primary interactions between the software and its users (Passengers and Fleet Operators) and its autonomous management logic. These functions are designed to satisfy the project goals.

User Management and Authentication

- **[R1] Registration and Login:** The system shall allow Passengers to create and manage personal accounts and allow both Passengers and Fleet Operators to securely authenticate into their respective interfaces.
- **[R2] Profile Management:** Users shall be able to update their personal information and credentials.

Ride Management

- **[R3] Immediate Ride Request:** The system shall allow Passengers to request a robotaxi for immediate pickup by specifying a destination.
- **[R4] Advance Booking:** The system shall allow Passengers to schedule a ride for a future date and time, managing availability to avoid overlapping assignments.
- **[R5] Automated Vehicle Allocation:** The system shall automatically assign an available robotaxi to a request based on the allocation strategy currently selected by the Fleet Operator.
- **[R6] Ride Status Notifications:** The system shall provide real-time updates to Passengers regarding vehicle assignment, ETA, arrival at the pickup point, and ride completion.

Fleet Supervision and Control

- **[R7] Real-time Fleet Monitoring:** The system shall provide Fleet Operators with a live overview of the entire fleet, displaying the position and state (e.g., available, assigned, maintenance) of every robotaxi.
- **[R8] Runtime Strategy Selection:** The system shall allow Fleet Operators to switch the active vehicle allocation policy (e.g., Nearest Available vs. Minimum ETA) at runtime without service interruption.
- **[R9] Maintenance Management:** The system shall allow robotaxis to be marked as unavailable for maintenance and prevent their assignment to rides during such periods.

Optimization and Rebalancing

- **[R10] Demand Analysis:** The system shall process data from external sources to identify high-demand zones in the city.
- **[R11] Proactive Fleet Rebalancing:** The system shall suggest or automatically trigger the repositioning of idle robotaxis toward high-demand areas to reduce future passenger waiting times.

Dynamic Allocation Strategy Management

To minimize customer wait times and optimize battery usage, the system manages the vehicle allocation strategy through two distinct operational modes: Auto and Manual.

- **[R12] Auto Mode (Default):** The system dynamically manages the active allocation strategy based on real-time traffic data.

  - The default strategy is Nearest Available Vehicle to optimize travel distance and battery consumption.
  - If traffic conditions reach a Medium threshold, the system alerts the Fleet Operator, suggesting a switch to the Minimum Estimated Time of Arrival (ETA) strategy.
  - If traffic conditions reach a High threshold, the system automatically switches the active strategy to Minimum ETA to efficiently handle the queue of requests.
  - To prevent a ping-pong effect, the system will automatically revert to the Nearest Available Vehicle strategy only when traffic conditions drop back to the Low threshold.
- **[R13] Manual Mode (Override):** The Fleet Operator retains ultimate control over the system.
  - If the Operator manually selects a specific allocation strategy, overriding the automated choice, the system immediately transitions to Manual Mode.
  - In Manual Mode, all automated strategy switches are suspended. The system will strictly follow the Operator's chosen strategy regardless of traffic changes.
  - The system remains in Manual Mode until the Operator explicitly re-enables Auto Mode via the dashboard.

2.5 Nonfunctional Aspects

The nonfunctional requirements (NFRs) specify the system's quality attributes, focusing on performance, reliability, usability, and maintainability. These aspects are critical to ensure that the ROAd platform operates efficiently in a dynamic urban environment.

Performance and Scalability

- **[NFR1] Concurrency Handling:** The system shall be capable of handling multiple concurrent ride requests and data updates from various users and robotaxis without performance degradation.
- **[NFR2] Real-time Responsiveness:** The notification system (based on the Observer pattern) shall deliver state-change updates to Passengers and Fleet Operators in near real-time to guarantee an accurate representation of the vehicle's status.
- **[NFR3] Scalability:** The centralized software architecture must be scalable to accommodate a growing fleet of robotaxis and an increasing number of registered passengers, supporting the rapid evolution of urban mobility.

Reliability and Consistency

- **[NFR4] Data Consistency:** The system must enforce strict database constraints to guarantee that vehicle availability timelines are accurately maintained, unequivocally preventing conflicting assignments of the same robotaxi to overlapping rides or bookings.

- **[NFR5] Robustness:** The system shall continuously track vehicle states explicitly via a State pattern, ensuring that transitions (e.g., from available to assigned or maintenance) are handled without reaching invalid or inconsistent states.

Usability

- **[NFR6] Ease of Use:** The ROAd interfaces, both the mobile application for Passengers and the web graphical user interface for Fleet Operators, shall be intelligent and easy-to-use, minimizing the learning curve required to request rides or monitor the fleet.

Maintainability and Extensibility

- **[NFR7] Extensibility of Allocation Logic:** The vehicle allocation logic must be implemented through a Strategy pattern, ensuring that it remains straightforward to introduce and deploy additional allocation algorithms in the future without altering the core system architecture.
- **[NFR8] Separation of Concerns:** The project architecture shall maintain a strict separation between its core management logic (user accounts, concurrent requests, state tracking) and external dependencies (autonomous driving, payment gateways, mapping services).

Reliability and Consistency

- **[NFR9] Stability of Auto-Switching:** The system shall implement hysteresis logic to prevent frequent and rapid oscillations between allocation strategies (the "ping-pong effect") when traffic levels fluctuate around the defined thresholds. Specifically, the system shall only revert to the Nearest Available strategy when traffic drops to the Low threshold, ensuring operational stability.
- **[NFR10] Priority of Human Intervention:** A manual command from the Fleet Operator must always take absolute precedence over the automated allocation logic. The system shall ensure an immediate and atomic transition to Manual Mode upon operator intervention, suspending all automated triggers to guarantee human-in-the-loop control.

2.6 Assumptions, Dependencies and Constraints

To clearly define the boundaries of the ROAd software architecture, several domain assumptions, external dependencies, and system constraints are established.

Domain Assumptions

- **[D1] Vehicle Autonomy:** It is assumed that the robotaxis are fully capable of autonomous driving and safe navigation within the urban environment. The core system does not handle the physical control or sensor data processing of the vehicles.
- **[D2] Connectivity:** It is assumed that both the robotaxis and the users (Passengers and Fleet Operators) maintain a reliable network connection to transmit GPS

positions, state changes, and ride requests to the centralized software system.

- **[D3] User Behavior:** It is assumed that passengers will be present at the specified pickup location at the time of the robotaxi's arrival.

Dependencies

- **External Mapping Services:** The system relies on third-party mapping providers to obtain geographical information, compute routes, calculate distances, and estimate travel times (ETAs). Crucially, the system also strictly depends on these services to retrieve real-time traffic level data (categorized as Low, Medium, or High). This real-time traffic data is the fundamental input required for the system to evaluate conditions and execute automated allocation strategy switching while operating in Auto Mode. Dynamic fleet rebalancing feature depends on external data sources or historical databases to provide the necessary information to identify expected high-demand zones.
- **Payment Gateways:** The processing of financial transactions and ride fares is treated as an external dependency and is not implemented as part of the core ROAd system.

Constraints

- **[C1] Non-Overlapping Assignments:** The system must strictly enforce database constraints to manage vehicle availability timelines, absolutely preventing conflicting assignments where the same robotaxi is scheduled for overlapping immediate rides or advance bookings.
- **[C2] Scope Limitation:** The focus of the project is strictly constrained to the software architecture, including managing user accounts, handling concurrent requests, tracking vehicle state transitions, and routing logic. The actual autonomous driving mechanics and payment processing are explicitly excluded from the implementation.

Chapter 3

Additional Models

3.1 Requirements-level Sequence Diagrams

3.2 Finite State Machines

Chapter 4

References